Supplementary Material

Casimir-cavity-induced conductance changes

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Device fabrication

In this work we fabricated metal-insulator-metal (MIM) devices, processed either by a germanium shadow-mask (GSM) process or standard photolithographic techniques.

GERMANIUM SHADOW-MASK DEVICES

The small feature sizes of the GSM devices, with critical dimensions on the order of hundreds of nanometers, are accomplished by using angled depositions underneath a germanium bridge that is supported by a polymethyl methacrylate (PMMA) under-layer, shown below in Fig. S1. This is a technique that has been employed in our group previously to make small area MIM diode junctions [1].

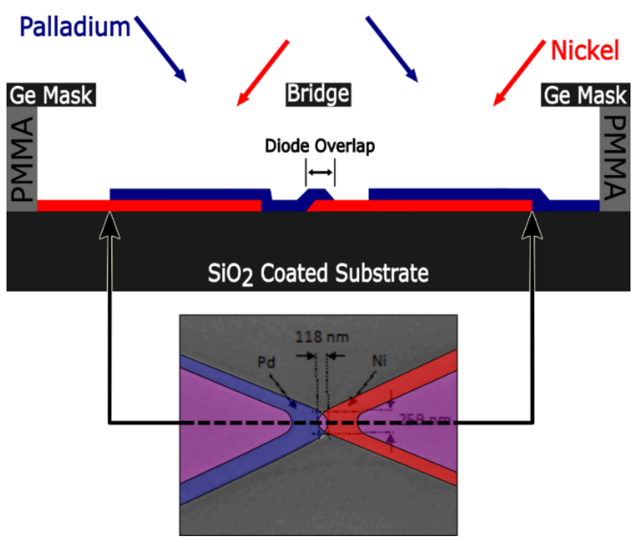

FIG S1: Cross-sectional view of the GSM process with materials deposited under the bridge.

To begin this process, a thermally oxidized silicon wafer with 300 nm of SiO_2 is spin coated with 4% PMMA in anisole at 1960 RPM for 45 seconds followed by a 15-minute curing bake at 180°C, resulting in a 260 nm thick PMMA coating. Then 60 nm of germanium is thermally evaporated onto the PMMA at a rate of 5 Å/sec. The germanium is then patterned, using a DUV stepper at University of California, Santa Barbara NanoFabrication Facility. This created the opening for the MIM depositions yielding a bridge that is 250 nm wide and 300 nm long. Once patterned, germanium is etched using an inductively coupled plasma (ICP) reactive ion etch (RIE) system with $CHF_3/CF_4/O_2$ gasses at a pressure of 0.5 Pa for 135 seconds. ICP power is 75 W with a RF substrate bias of 75 W. This step completely etches through germanium and etches ~30 nm of underlying PMMA. Once the germanium is etched, a RIE system is used to completely etch the remaining PMMA in O_2 with a power of 100 W at a pressure of 350 mTorr for 6 minutes. An illustrated top view of the patterned germanium shadow mask is shown below in Fig. S2. The leads and pads used in GSM devices are similar to those shown for the large area devices shown in Fig. S3.

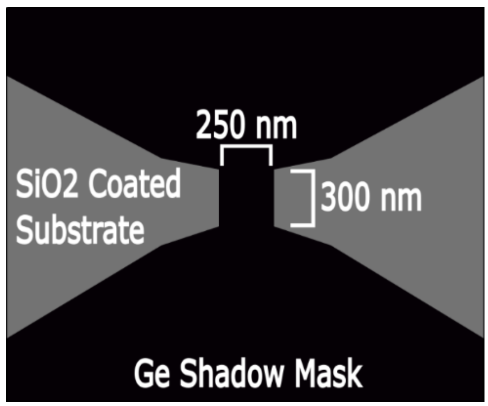

FIG S2: Top view of a GSM prior to any material depositions.

Once the GSM is complete, the MIM structure is formed using angled evaporations, as depicted in Fig. S1. First, 50 nm of Ni is thermally evaporated (1.5 Å/sec) at an angle of 39.3°. Next, 1.3 nm of non-stoichiometric Al_2O_3 is RF sputtered at 75 W with 30 sccm O_2 and 20 sccm Ar using an Al_2O_3 target. Since this is not an in-situ process, native oxide growth on Ni base metal is inevitable. Based on our electrical measurements (linear current-voltage characteristics and <30 Ω zero-bias differential resistance) and simulations on Ni/native NiO_x/Pd structures, we extracted a NiO_x thicknesses of 0.6-1 nm for effective barrier heights of 0.06-0.08 eV [2]. The barrier height was calculated by performing Fowler-Nordheim analysis on a low resistance (~100 Ω) device with nonlinear current-voltage characteristics. Hence, the total effective insulator thickness for the Al_2O_3/NiO_x combination is 2.3 nm. The MIM structure is completed with 11 nm of thermally evaporated Pd (1.3 Å/sec), angled at 39.3° opposite to the direction of Ni evaporation. Taking into account the angle of evaporation, the thickness of Ni base metal is 38 nm, and the thickness of Pd upper electrode is 8.3 nm. PMMA and Ge is then lifted off using acetone. The final step in the process is to defined the Casimir cavity over the MIM junction with an area of 7 μm^2 using standard photolithographic techniques via an etch-through process. To accomplish this, PMMA is spun on the wafer, with thicknesses ranging from 30 nm to 1100 nm, followed by a thermally evaporated 150 nm thick aluminum layer at a rate of 6 Å/sec. Finally, the aluminum and PMMA are wet etched and dry etched, respectively.

LARGE AREA DEVCES

Large area devices are made using standard photolithographic techniques on a thermally oxidized silicon wafer with 300 nm SiO_2. NR9 resist is patterned for the bottom metal and 50 nm of Ni is then thermally evaporated at a rate of 2.7 Å/sec and lifted off in heated (60 °C) RR5. As with the GSM devices, the native oxide growth on exposed Ni is inevitable, and hence we measured a native NiO_x thickness of 2.3 nm by variable angle spectroscopic ellipsometry (VASE), described below. This is thicker than for GSM structures as the junctions in GSM structures is partially protected by the germanium bridge. A blanket layer of 1.3 nm, non-stoichiometric Al_2O_3 is RF sputtered at 75 W with 30 sccm O_2 and 20 sccm Ar using an Al_2O_3 target. NR9 is patterned for the upper electrode, Pd which is evaporated with thicknesses from 8.7 nm to 24 nm (1.5 Å/sec). Unfortunately, descumming is not an option because it would induce additional oxide growth, and thus a monolayer (0.4 nm) of NR9 resist is not removed prior to Pd deposition. The upper electrode Pd is patterned such that a square area overlaps on top of the Ni pad ranging in size from 6.25 μm^2 to 10000 μm^2. Palladium is lifted off once more with RR5 at 60° C. Once the MIM device is complete, NR9 is patterned for the definition of the Casimir cavity. This pattern is centered over the MIM junction and sized so that it overshadows the MIM device. Then the cavity oxide and mirror are deposited, with the cavity oxide typically 12 nm of SiO_2 deposited via RF sputtering SiO_2 target at 50 W with 30 sccm O_2 and 20 sccm Ar gasses. The cavity pattern is formed by liftoff in RR5 heated to 60 °C, resulting in the device depicted in FIG S3.

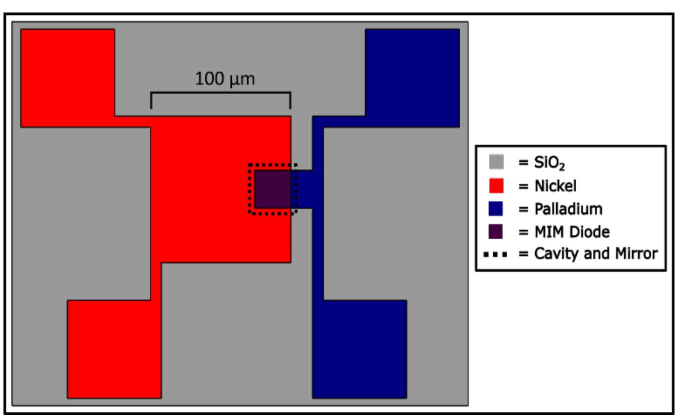

FIG S3: Top view of a standard photolithography device with the Casimir cavity.

Electrical Measurements

Once the MIM structures were fabricated, current-voltage (I(V)) measurements were performed with a precision of 1 μV and <1 nA in voltage and current, respectively. We used a four-point probe configuration to perform I(V) measurements so that the active MIM junction could be isolated from parasitic lead resistance. A high precision Keithley 2612 source meter (calibrated to NIST standards) was used to source either voltage or current across two pads and a HP 3478A digital multimeter (DMM) was used to measure the voltage drop across the junction. We incorporated a mercury switch to short out all the four contact pads during probe manipulation to prevent static discharge from damaging the MIM junction. Although, sourcing a voltage and measuring the current using the source meter is the standard and most straightforward approach of taking I(V) measurements, this technique is problematic when measuring currents on the order of 1 nA through a device having a resistance less than 1 MΩ [See: Keithley: Series 2600 system sourcemeter, User's manual, United States America (2006)]. This is because the precision of the source meter in voltage sourcing mode is low, as the voltage offset is large (applied bias + 0.02% + 250 μV). For instance, if we are measuring a device with a differential resistance of 1 kΩ with 1 nA precision, the corresponding voltage is 1 μV, which is within the error. On the other hand, if we measure a 1 MΩ device in 1 nA precision, the corresponding voltage is 1 mV, and thus the measurement can be done without significant error. Using this standard measurement setup, we tested reference MIM structures with differential resistances of ~200 Ω and resistors with matching resistances. We observed an offset current of ~60 nA and an offset voltage of ~10 μV even after applying the current reversal method to eliminate any systematic contribution, such as a thermoelectric offset due to the temperature difference between the source meter and the probes [See: C. Miller, Techniques for Reducing Resistance Measurement Uncertainty: DC Current Reversals vs. Classic Offset Compensation, White paper, Keithley Instruments, Inc. (2004)]. In using the current reversal method, we performed two measurements with currents of opposite polarity, one when the base metal is grounded, another when the upper electrode was grounded. Then we subtracted the current for the scenario when base electrode was grounded from the current when upper electrode is grounded. Hence, the effect due to thermoelectric potentials was eliminated.

Alternatively, sourcing a current and measuring voltage is much more precise (applied current ± 0.06% + 100 pA) according to manufacturer specifications [See: Keithley: Series 2600 system sourcemeter, User's manual, United States America (2006)]. We used the same four-point-probe configuration, two probes to source current and measure voltage using the source meter and the other two probes connected to the DMM to measure the voltage drop across the junction. We verified the precision of measurement configuration by performing I(V) sweeps on four types reference MIM structures:

1) With a 1100 nm thick Casmir cavity
2) With only the cavity dielectric (without the Al reflector)
3) Without the cavity dielectric, but annealed at 180° C for 15 minutes to match processing conditions that the device undergo when defining the cavity
4) Completed, with resistances varying from 120 Ω to 6100 Ω.

We also performed I(V) measurements on resistors with matching resistance values. However, due to the use of two different metals having different work functions in MIM junctions, a contact potential can be created across the two metals which does not occur with resistors, and hence we used MIM devices as the reference/control devices. These measurements were carried out using a manufacturer-calibrated Keithley source meter. As shown in Fig. S4(a), it is evident that the current and voltage offset values are <1 nA and <1 μV, respectively for reference MIM devices with no Casimir cavities after applying the offset correction using the current reversal method.

We used both voltage-sourcing and current-sourcing measuring techniques depending on the differential resistance of the device under test and the precision required for the measurement.

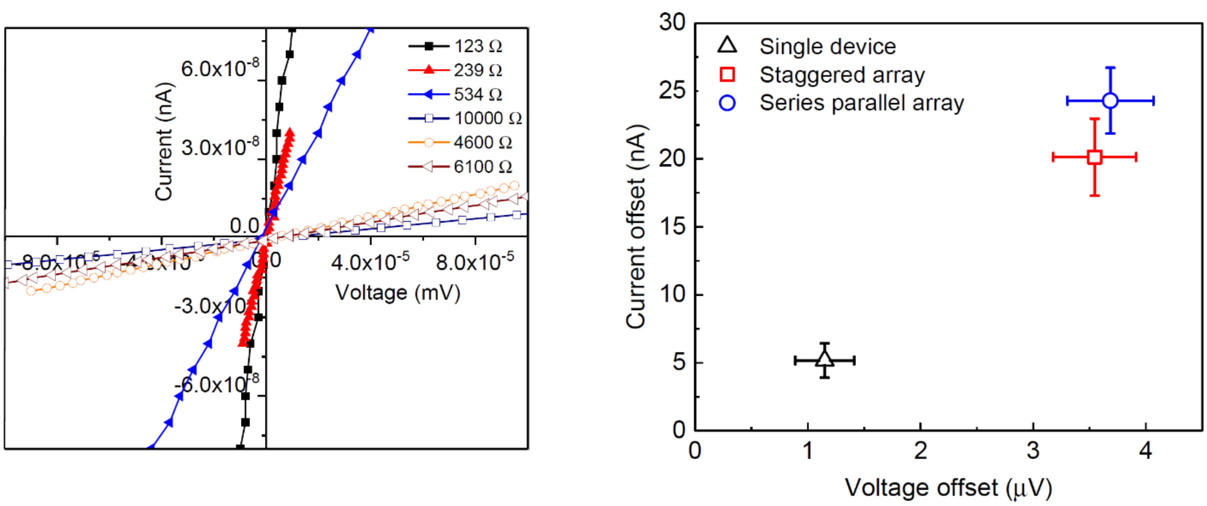

FIG S4: (a) Current-voltage characteristics on reference MIM devices, which have no Casimir cavities with different resistance values. (b) Current and voltage offsets for individual GSM device and 4 x 4 arrays produced under the same conditions; every data point corresponds to an average over 6 devices.

As discussed in the main text, the main objective here is to eliminate sources of error that could produce artifacts in measurements. We accomplished this in the past for MIM structures and resistors without Casimir cavities. However, when we formed the Casimir cavity on top of the MIM devices, I(V) characteristics did not pass through the origin, as shown in Fig. 5 of the main text. Our first attempt was to understand if this offset is generated by a thermal effect, as the offset voltage magnitudes are small (~6 μV). First, we measured the temperature on the substrate stage (21.6 ° C) and the ambient (23.4 ° C). Since, Ni is our base electrode, this could be colder than the Pd upper electrode. If we consider the bulk Seebeck coefficients for metals (Ni = -18 μV/° C, Pd = -9 μV/° C) or the insulator (> 100 μV/° C), a temperature differential of < 1° C is sufficient to produce these voltage magnitudes. However, since our metals and insulators are very thin, it is not possible to calculate the voltage for a given temperature differential by using bulk Seebeck coefficients, as they can vary significantly depending on material and electronic properties for a given material. Therefore, a series of measurements were performed with the goal of observing a change in offset voltage values by varying the stage and ambient temperatures. A heat gun was used to elevate the temperature of the stage up to 24.8 ° C (without varying the temperature of ambient). We then measured the voltage offset which did not vary over a duration of 20 minutes from the value it had before the heat gun was applied. We performed the same experiment by increasing the temperature of the ambient (without varying the temperature of the stage), and again observed no difference in offset voltage. In a third experiment, we cooled down the temperature of the ambient to 21 ° C using a portable air conditioner, and once more no difference in offset voltage was observed. Hence, the temperature of the base and upper electrodes remains very close to each other, and the observed offset voltage cannot be generated solely due to a thermal effect.

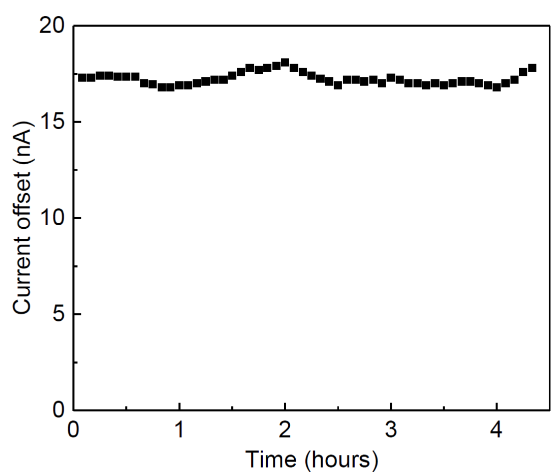

FIG S5: Current offset with no applied voltage as a function of time for a 38 nm Ni/1.7 nm Al_2O_3+NiO_x/8.3 nm Pd/33 nm PMMA/150 nm Al device.

Another artifact check was performed by making measurements on arrays of GSM Casimir photoinjectors configured as 16 devices with pairs of devices in series and parallel, so that each array contains four sets of devices in parallel and also in series. If the results obtained with single devices are accurate, then the offsets in both voltage and current should scale with the number of devices in the array, in this case, a factor of four. As can be seen in Fig. S4(b), the current and voltage offset for the arrays are, in fact, a factor of four larger than those for a single device. This further confirms that the offset is not a measurement artifact.

Another possibility is the offset could be due to some sort of hysteresis effect or transient response caused by charge buildup at thin metal-insulator interfaces. If so, the measured current should decay over time. To test for this, the current with no applied voltage was measured as a function of time for over 4 hours. No decay in the offset current was observed, as can be seen in Fig. S5. This shows that the offset is not due to a charging effect, and this is further confirmed by hysteresis measurement, in which no hysteresis was observed.

With the goal of investigating whether variations in the conductance are due to the full Casimir cavity or just the presence of the dielectric, or some special property of PMMA rather than the presence of a Casimir cavity, we performed a study on devices at sequential stages of formation with cavities comprising either PMMA or SiO_2 transparent dielectrics, as can be seen in Fig S6. The purpose for including an annealed MIM structure is to simulate the thermal effects of the processing used to pattern the cavity dielectric and mirror. It is evident that the differential conductance values of the completed device with mirrors are significantly higher compared to devices without a Casimir cavity or with a cavity dielectric but without mirrors, supporting the Casimir cavity model for the conductance change, and showing that measurement and processing anomalies cannot explain the results.

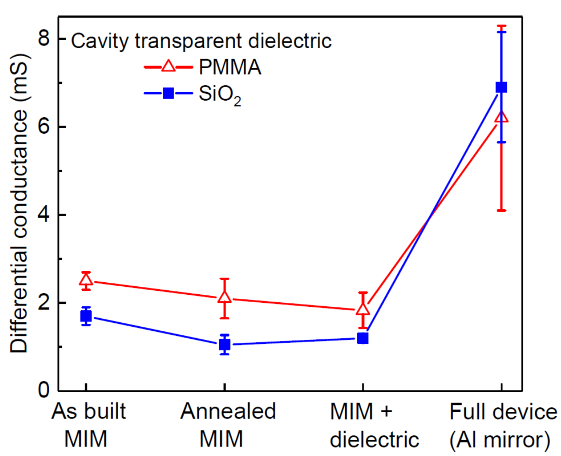

FIG S6: Differential conductance for devices with and without Casimir cavity. The devices with PMMA transparent dielectric (36 nm) had an effective transport insulator of 1.7 nm (native NiO_x+Al_2O_3) whereas devices with SiO_2 transparent dielectric (50 nm) had an effective transport insulator of 1.9 nm (native NiO_x+Al_2O_3). The mirror material for all the devices with Casimir cavities was 150 nm of evaporated Al.

Because of its high near-UV reflectivity, we generally use Al as the mirror material. To understand its impact on the results, we compared otherwise identical photoinjector devices having Al mirrors to ones with Ag. The device parameters are 38 nm Ni base electrode, 1.9 nm $NiO+Al_2O_3$ insulator, 8.3 nm Pd upper electrode, 50 nm SiO_2 cavity, and 150 nm Al, or 150 nm Ag + 3 nm Ti mirrors. The differential conductance with Al and Ag mirrors are 3.1 +/- 0.3 mS and 3.2 +/- 1.5 mS, respectively, even though the reflectivity of Ag drops substantially for wavelengths below 450 nm. The similar conductance for the two types of devices suggests that the short wavelengths do not produce photoinjected carriers efficiently. That conclusion is consistent with the transport of high energy carriers being limited by interband transitions [3].

The probe station that we were using to make measurements has a high precision motorized translational stage. Thus, the magnetic field strength was significant around the stage. We performed high precision I(V) measurements on a different probe station where there was no motorized stage and observed no difference in measurements, i.e., the same offset magnitudes were observed for both current and voltage. To see if the ambient lighting can influence I(V) characteristics, we performed hundreds of I(V) measurements with and without any light, and we took extra measures to cover the entire probe station with a blackout cloth purchased from Thor labs. Again, there were no differences in I(V) characteristics for MIM Casimir photoinjectors. We have thus far been unable to eliminate the anomalous voltage and current offsets or find an artifact that would explain them. We anticipate reporting the source of these anomalies in a future publication.

Variable angle spectroscopic ellipsometry (VASE) measurements for the extraction of thickness and optical properties

To characterize the optical properties and thicknesses of the insulators and transparent cavity filler materials we have carried out a series of VASE measurements. The UV-Vis-NIR VASE measurements were performed using a spectral range from 0.7 to 6.4 eV (corresponding to a wavelength range λ = 194 nm to 1700 nm), and angles of incidence of 55°–75° with steps of 10° to improve the accuracy of the measurement. A Woollam M2000UI VASE tool (located at University of California, Santa Barbara) and completeEASE® software were used to measure and analyze thin films deposited or grown on Si substrates. The measured data was imported to completeEASE®, which generated a fit for measured experimental data by using a Cauchy layer in the transparent region. Then we extracted the thickness and optical properties (refractive index and extinction coefficient) of a given thin film using the generated model as explained in [See: Débora Gonçalves and Eugene A. Irene, Fundamentals and applications of spectroscopic ellipsometry, Química Nova 25(5) 794-800 (2002)]. The extracted parameters are summarized in Table 1.

Table 1: Summary of measured variable angle spectroscopic ellipsometry analysis.

Material	Refractive index at λ = 300 nm	Band gap (eV)	Thickness (nm)
Al_2O_3	1.72	6.20	2.86
SiO_2	1.49	> 6.40 eV	1.56
PMMA	1.52	3.47	33.13
Pd			8.32

We made sure to place a witness sample when depositing each thin film to check whether we have deposited the desired thickness under the assumption that the calibrated deposition rates were still valid. Also, we used the extracted band gap and Fowler-Nordheim analysis to determine conduction band and valence band offset values at the interface of Pd/Al_2O_3. The conduction band offset (barrier height) of 0.3 eV was extracted from Fowler-Nordheim analysis, yielding a valence band offset of 5.9 eV at the interface of Pd/Al_2O_3, hence, it is clear that the contribution from hot holes is weaker than that of hot electrons.

References

[1] B. Pelz and G. Moddel, Demonstration of distributed capacitance compensation in a metal-insulator-metal infrared rectenna incorporating a travelling-wave diode, Journal of Applied Physics **125** 234502-1 – 234502-7 (2018).

[2] A. Weerakkody, A. Belkadi, and G. Moddel, unpublished.

[3] G. Tagliabue, A. S. Jermyn, R. Sundararaman, A. J. Welch, J. S. DuChene, R. Pala, A. R. Davoyan, P. Narang, and H. A. Atwater, Quantifying the role of surface plasmon excitation and hot carrier transport in plasmonic devices, Nature communications **9** 1-8 (2018)